

Data Exercise

Using Microsoft Excel in a more effective and efficient way

Outline

1. Mindset
2. Livestock not pets
3. Project-oriented workflow

Mindset

As our skills as data analysts grow, the surrounding systems and infrastructure become crucial for ensuring the reproducibility and long-term preservation of our work. However, we often lack formal training or mentorship in managing such systems, leading us to either get lost amidst this technology or that we begin to explore these possibilities ourselves.

Mindset

- With this short course, we hope to help you gracefully fall into this gap.
- Don't fret over past mistakes, but raise the bar for new work. Small but meaningful incremental changes add up over time, transforming your **data quality of life**.

Livestock not pets

Think of your data processes as livestock and not pets.

Spreadsheets: save a source for your written functions

- Use a text editor (rather than a word processor)
 - For Windows: Notepad++ - <https://notepad-plus-plus.org/>
 - For Mac: CodeEdit - <https://www.codeedit.app/>
- Create a text file to associate with every spreadsheet project that you are working on.
- Save the text file within the same directory as the associated spreadsheet.

Project-oriented workflow

Organise work into projects.

File system discipline

- Put all the files related to a single project in a designated folder.
- This applies to data, code, figures, notes, etc.
- Depending on project complexity, you might enforce further organization into subfolders.

File naming

File organisation and naming are powerful weapons against chaos.

Three principles for file names

Name ▼	Size	Modified	
 analysis for KA 1 PLAs.xlsx.gpg	412.6 kB	03:15	☆
 Attainment result term 3.doc.gpg	11.4 kB	03:15	☆
 data analysis PLAs WSI 2023.xlsx.gpg	224.7 kB	03:15	☆
 data for KA 1 PGAp wsi 2023.xlsx.gpg	34.6 kB	03:15	☆
 P6 National Exam Results 2022.pptx.gpg	845.5 kB	03:15	☆
 S4_2021_option.xlsx.gpg	30.5 kB	Fri	☆
 subject Report Term 1 2023.docx.gpg	19.4 kB	03:15	☆
 TERM 1 2023 RESULTS.xlsx.gpg	46.0 kB	03:15	☆

- machine-readable
- human-readable
- plays well with default ordering

Machine-readable

- avoid spaces, punctuation, accented, characters, case sensitivity.
- deliberate use of delimiters/space-holders - "_" or "-"
 - general rule is that "_" to delimit units of metadata while "-" to delimit words so that they are more readable.

Machine-readable

- easy to search for files later
- easy to narrow file lists based on names
- easy to extract information from file names

Human-readable

- name contains info on content.
- use the concept of a **slug** used in URLs.

Human-readable

Easy to figure out what something is based on its name

Plays well with default ordering

- put something numeric first
- use ISO 8601 standard for dates
- add leading zeros to numbers

Three principles for file names

- easy to implement now
- payoffs accumulate as your skills evolve and projects get more complex

Exercise

The `fem` data is a dataset from 118 female students consulting at the university mental health/psychiatry clinic.

Data dictionary

ID	Patient ID
AGE	Age in years
IQ	IQ score
ANX	Anxiety (1=none, 2=mild, 3=moderate, 4=severe)
DEP	Depression (1=none, 2=mild, 3=moderate or severe)
SLP	Sleeping normally (1=yes, 2=no)
SEX	Lost interest in sex (1=yes, 2=no)
LIFE	Considered suicide (1=yes, 2=no)
WT	Weight change (kg) in previous 6 months